5G RAN

SA Mobility Management in Inactive Mode Feature Parameter Description

Issue Draft A

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue does not introduce any changes to 5G RAN9.1 03 (2025-08-30).

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-031102	Inactive State	**4 Mobility Management in Inactive Mode**

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Mobility management in inactive mode	UEs in inactive mode are not supported in NSA networking. Mobility management in inactive mode is supported only in SA networking.	**4 Mobility Management in Inactive Mode**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Mobility management in inactive mode	Supported only in low frequency bands	**4 Mobility Management in Inactive Mode**

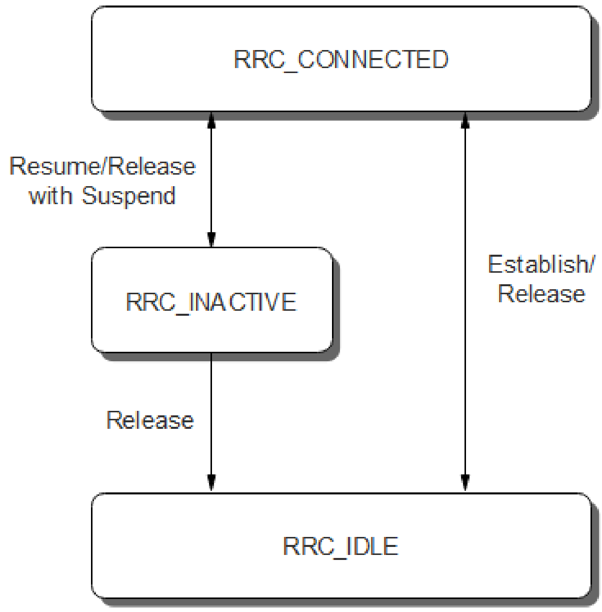
3 Overview

Mobility management is a basic function used to ensure that services stay uninterrupted as UEs move through mobile networks.

3.1 RRC State

In a 5G network, **a UE at any time is in one of the following radio resource control (RRC) states: RRC_CONNECTED, RRC_IDLE, and RRC_INACTIVE states**, as presented in **Figure 3-1**.

Figure 3-1 RRC states



- RRC_CONNECTED: An RRC connection is established between the UE and the gNodeB.
- RRC_IDLE: No RRC connection is established between the UE and the gNodeB.
- RRC_INACTIVE: In this state, the UE suspends data processing. However, the gNodeB still maintains the UE's context information, and as a result, the core network considers that the UE remains in the CM-CONNECTED state (as if it

were still in the RRC_CONNECTED state). When data transmission is required, the UE can quickly transit to the RRC_CONNECTED state.

NOTE

RRC connection release is performed for transition from RRC_CONNECTED to RRC_IDLE or from RRC_CONNECTED to RRC_INACTIVE. The state (RRC_IDLE or RRC_INACTIVE) that a UE has entered can be checked based on whether *suspendConfig* is included in the RRCRelease message. If *suspendConfig* is included in the RRCRelease message, the UE has entered the RRC_INACTIVE state. If *suspendConfig* is not included in the RRCRelease message, the UE has entered the RRC_IDLE state.

For details about the RRC states, see section 4.2.1 "UE states and state transitions including inter RAT" in 3GPP TS 38.331 V15.5.1.

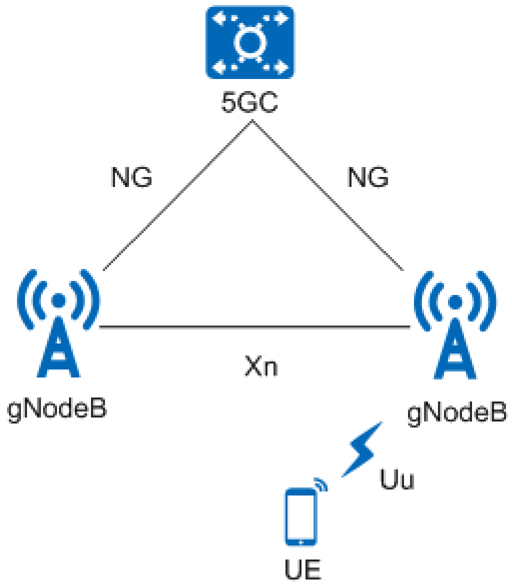
3.2 Mobility Management Scenarios

5G supports standalone (SA) networking and non-standalone (NSA) networking. Based on networking scenarios, mobility management is categorized into SA mobility management, NSA mobility management, and NSA and SA hybrid mobility management. **This document focuses on functions related to mobility management for UEs in inactive mode in SA networking.**

SA Mobility Management

SA networking uses an end-to-end 5G network architecture, where the terminals, base stations, and core network comply with 5G specifications. Figure 3-2 shows SA networking. For details about SA networking, see *5G Networking and Signaling*.

Figure 3-2 SA networking (Option 2 as an example)



In SA networking, mobility management is classified into intra-RAT mobility management and inter-RAT mobility management, depending on whether a serving cell changes between two networks of different radio access technologies (RATs). For details about SA inter-RAT mobility management, see *Interoperability*

Between E-UTRAN and NG-RAN in Connected Mode, *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*, and *NSA-SA Selection Based on User Experience*.

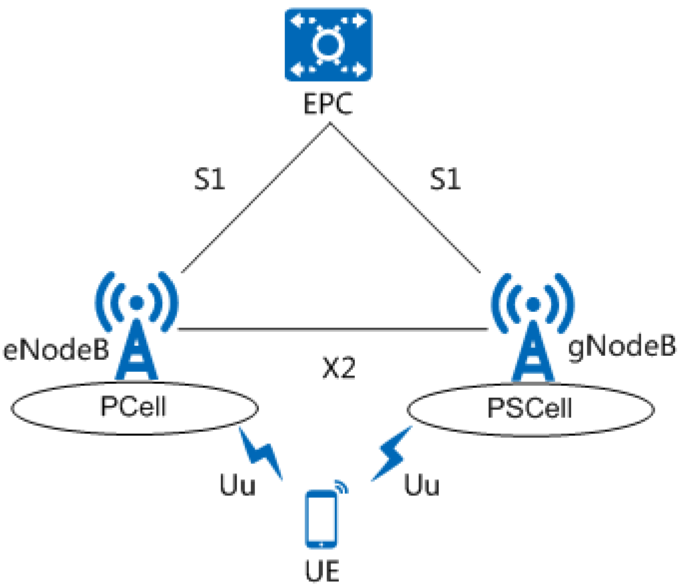
Based on the RRC states of UEs, intra-RAT mobility management in SA networking is categorized as follows:

- Mobility management in connected mode (RRC_CONNECTED state). For details, see *SA Mobility Management in Connected Mode*.
- Mobility management in idle mode (RRC_IDLE state). For details, see *SA Mobility Management in Idle Mode*.
- Mobility management in inactive mode (RRC_INACTIVE state), which is described in this document.

NSA Mobility Management

In NSA networking, terminals supporting E-UTRA-NR dual connectivity (EN-DC) connect to both an LTE base station and a 5G New Radio (NR) base station, using the radio resources of both of the base stations for transmission. The cell served by the LTE base station (eNodeB) is the primary cell (PCell), and the cell served by the 5G NR base station (gNodeB) is the primary SCG cell (PSCell). **Figure 3-3** shows the NSA EN-DC networking. For details about NSA networking, see *EPC-based NSA Basics*.

Figure 3-3 NSA EN-DC networking (Option 3x as an example)



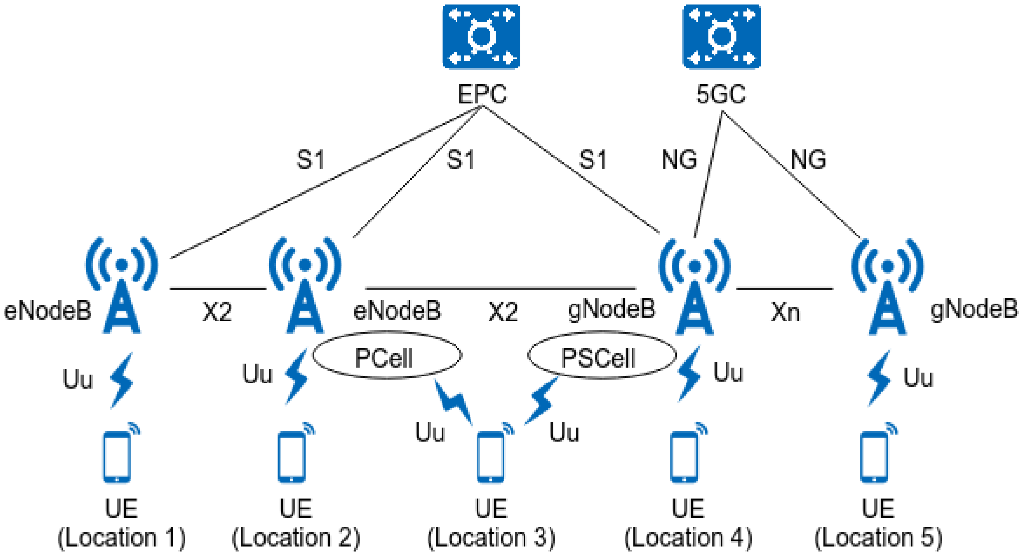
The RRC_INACTIVE state is not supported in NSA networking. Based on the RRC states of UEs, mobility management in NSA networking is categorized as follows:

- Mobility management in the RRC_CONNECTED state: UEs are connected to both the PCell and PSCell. Therefore, mobility management is categorized as PCell change and PSCell change. For details, see *NSA Mobility Management*.
- Mobility management in the RRC_IDLE state: UEs camp on LTE cells. For details, see *Idle Mode Management* in eRAN Feature Documentation.

NSA and SA Hybrid Mobility Management

NSA and SA hybrid networking marks a transition in the smooth evolution from NSA networking to SA networking, as shown in **Figure 3-4**.

Figure 3-4 NSA and SA hybrid networking



As shown in **Figure 3-4**, the following mobility scenarios occur in NSA and SA hybrid networking:

- When a UE moves from location 1 to location 2: The UE camps on an LTE cell at location 1 or location 2. In this scenario, intra-LTE mobility management is performed and the LTE mechanisms apply. For details, see *Idle Mode Management* and *Mobility Management in Connected Mode* in *eRAN Feature Documentation*.
- When a UE moves from location 2 to location 3: The UE camps on an LTE cell at location 2 and an NR cell in NSA networking at location 3. In this scenario, NSA mobility management is performed. For details, see *NSA Mobility Management*.
- When a UE moves from location 3 to location 4: The UE camps on an NR cell in NSA networking at location 3 and an NR cell in SA networking at location 4. In this scenario, interoperability between E-UTRAN and NG-RAN is performed. For details, see *NSA-SA Selection Based on User Experience*.
- When a UE moves from location 4 to location 5: The UE camps on an NR cell in SA networking at location 4 or location 5. In this scenario, SA mobility management is performed. If the UE is in connected mode during the movement, SA mobility management in connected mode is performed. For details, see *SA Mobility Management in Connected Mode*. If the UE is in idle mode during the movement, SA mobility management in idle mode is performed. For details, see *SA Mobility Management in Idle Mode*. If the UE is in inactive mode during the movement, SA mobility management in inactive mode is performed, as described in this document.
- When a UE directly moves from location 1 to location 5: The UE camps on an LTE cell at location 1 and an NR cell in SA networking at location 5. In this scenario, interoperability between E-UTRAN and NG-RAN is performed. For

details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode* and *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

4 Mobility Management in Inactive Mode

This chapter describes SA mobility management policies for UEs in inactive mode.

4.1 Principles

The RRC_INACTIVE state is introduced to NR. A UE in this state suspends data processing and, when moving inside a RAN-based Notification Area (RNA), does not need to exchange information with the gNodeB. (For details about RNA, see **4.1.1 RNA Introduction**.) As the gNodeB continues to maintain the UE's context information, the core network considers that the UE remains in the CM-CONNECTED state (as if it were still in the RRC_CONNECTED state). When data transmission is required, the UE can quickly transit to the RRC_CONNECTED state. **The UE in the RRC_INACTIVE state can maintain the similar level of power consumption as it did in the RRC_IDLE state while resuming data transmission within a short delay.**

The RRC_INACTIVE state is enabled by selecting the **RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch.*InactiveStrategySwitch*** parameter. Note that, for RedCap and eRedCap UEs, the RRC_INACTIVE state is enabled by selecting the **REDCAP_RRC_INACTIVE_SWITCH** option of the **NRCellAlgoSwitch.*InactiveStrategySwitch*** parameter. For details, see the "RedCap UE Inactivity Management" section in *RedCap*.

An RRC_CONNECTED UE enters the RRC_INACTIVE state when all of the following conditions are met:

- The switch for the RRC_INACTIVE state is turned on, and the UE supports this state.
- The no-data-transmission duration of the UE exceeds the value of **NRDUCellQciBearer**.***UeInactivityTimer***.

  For a UE performing composite services, the maximum value of **NRDUCellQciBearer**.***UeInactivityTimer*** configured among all QCIs of the UE is used in the preceding comparison. Such a UE has multiple radio bearers, each of which corresponds to a QCI configured with a specific value of **NRDUCellQciBearer**.***UeInactivityTimer***.

NOTE

If the no-data-transmission duration of an RRC_CONNECTED UE exceeds the value of **NRDUCellQciBearer**.***UeInactivityTimer***, the gNodeB releases the UE regardless of whether the UE supports the RRC_INACTIVE state. The UE then enters the RRC_IDLE or RRC_INACTIVE state depending on its capability.

An RRC_CONNECTED UE that supports the RRC_INACTIVE state can be released in advance so that the UE enters the RRC_INACTIVE state at an earlier time. The UE can quickly transit to the RRC_CONNECTED state when data transmission of services arrives, with little impact on services. In this context, Huawei introduces the RRC_INACTIVE UE power saving based on the inactivity timer function. This function uses the **NRDUCellQciBearer**.***InactiveUeInactivityTimer*** parameter to configure an inactivity timer specifically designed for UEs that can operate in the RRC_INACTIVE state. As a result, these UEs can be released earlier into the RRC_INACTIVE state, helping to achieve UE power saving. For details about the principles and configurations, see *UE Power Saving*.

- The RNA ID of the serving cell has been planned. That is, the **NRDUCell.*RanNotificationAreaId*** parameter is set to a valid value (not **65535**).

A UE in the RRC_INACTIVE state performs cell search, PLMN selection, cell selection, cell reselection, RNA update, and state transition during its movement.

- The mechanisms of cell search, PLMN selection, cell selection, and cell reselection are the same as those for a UE in the RRC_IDLE state. For details, see *SA Mobility Management in Idle Mode*.
- RNA update: This procedure helps acquire the accurate RNA of the moving UE. For details, see **4.1.2 RNA Update**.
- State transition: The UE changes its state to RRC_CONNECTED or RRC_IDLE in response to the change in service status. For details, see **4.1.3 State Transition from RRC_INACTIVE**.

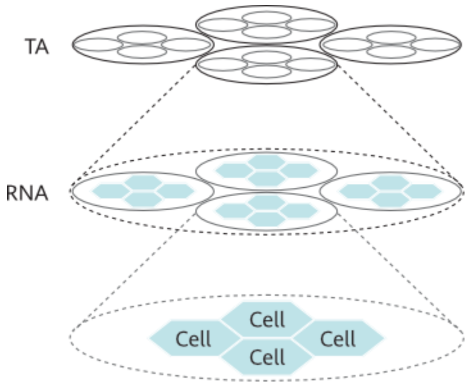
4.1.1 RNA Introduction

To locate a UE in the RRC_IDLE state, the core network sends paging messages in all cells of all tracking areas (TAs) in the tracking area identity (TAI) list allocated to the UE. This procedure is called 5GC paging. The resulting transmission overheads are obviously high as a plurality of paging messages are sent in cells where the target UE is not located.

To reduce transmission overheads of paging a UE in the RRC_INACTIVE state, the gNodeB sends paging messages at a smaller granularity (RNA) than TA. This procedure is called RAN paging, and RNA is managed by the gNodeB. RNA is short for RAN-based Notification Area.

As shown in **Figure 4-1**, **an RNA consists of one or more cells, and the cells in the same RNA must belong to the same TA**.

Figure 4-1 Relationship among cells, RNAs, and TAs

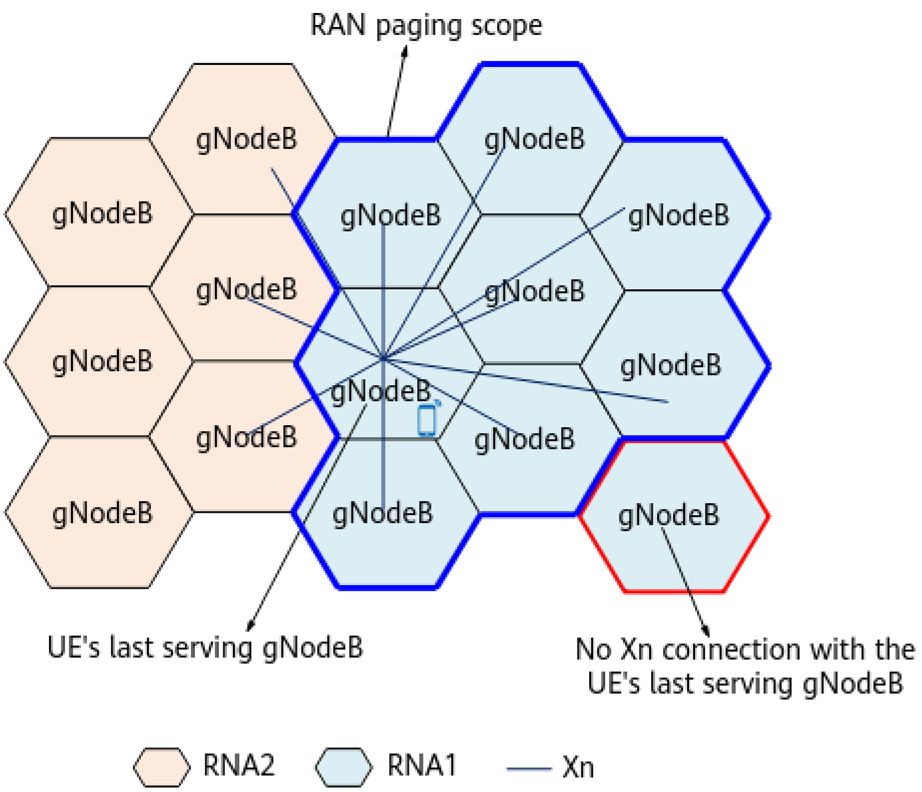


As illustrated in **Figure 4-2**, a UE belongs to RNA1. **In RAN paging, paging messages are sent to all external cells that are served by gNodeBs having Xn connections with the UE's last serving gNodeB and for which the RNA ID is RNA1.** That is, paging messages are sent to the area outlined in blue. The RNA IDs of external cells are sent to the serving cell over the Xn interface. Note that paging messages will not be sent to the external cells served by gNodeBs having no Xn connections with the UE's last serving gNodeB or for which the RNA ID is not RNA1. Therefore, Xn interfaces must be set up between gNodeBs in the same RNA, and proper RNA IDs must be configured for the external cells as planned.

NOTE

For details about RAN paging, see *5G Networking and Signaling*.

Figure 4-2 RAN paging



When a UE enters the RRC_INACTIVE state, the last serving gNodeB assigns an RNA ID (including a TAC and a RAN area code) to the UE.

In scenarios without multi-operator sharing, the RAN area code is specified by the **NRDUCell**.***RanNotificationAreaId*** parameter. In multi-operator sharing scenarios, the operator-specific RAN area code is specified by the **NRDUCellOp**.***RanNotificationAreaId*** parameter. If cell-list-based automatic RNA planning has been enabled, manual configuration of the RAN area code is not required. For details about cell-list-based automatic RNA planning, see *UE Power Saving*.

The UE obtains the RNA ID in either of the following ways:

- If the UE transits from the RRC_CONNECTED state to the RRC_INACTIVE state, it obtains the RNA ID of the current cell through the *RAN-AreaConfig* IE in the RRCRelease message sent from the base station.
- After cell reselection for the RRC_INACTIVE UE is complete, the RRC_INACTIVE UE obtains the RNA ID of the current cell from the *RAN area code* and *Tracking area code* IEs in SIB1. The RRC_INACTIVE UE then compares the RNA ID with the last-acquired RNA ID. If the two RNA IDs are different, an RNA update procedure is initiated and the RRC_INACTIVE UE sends an RRCResume message to notify the base station of the RNA ID of the current cell.

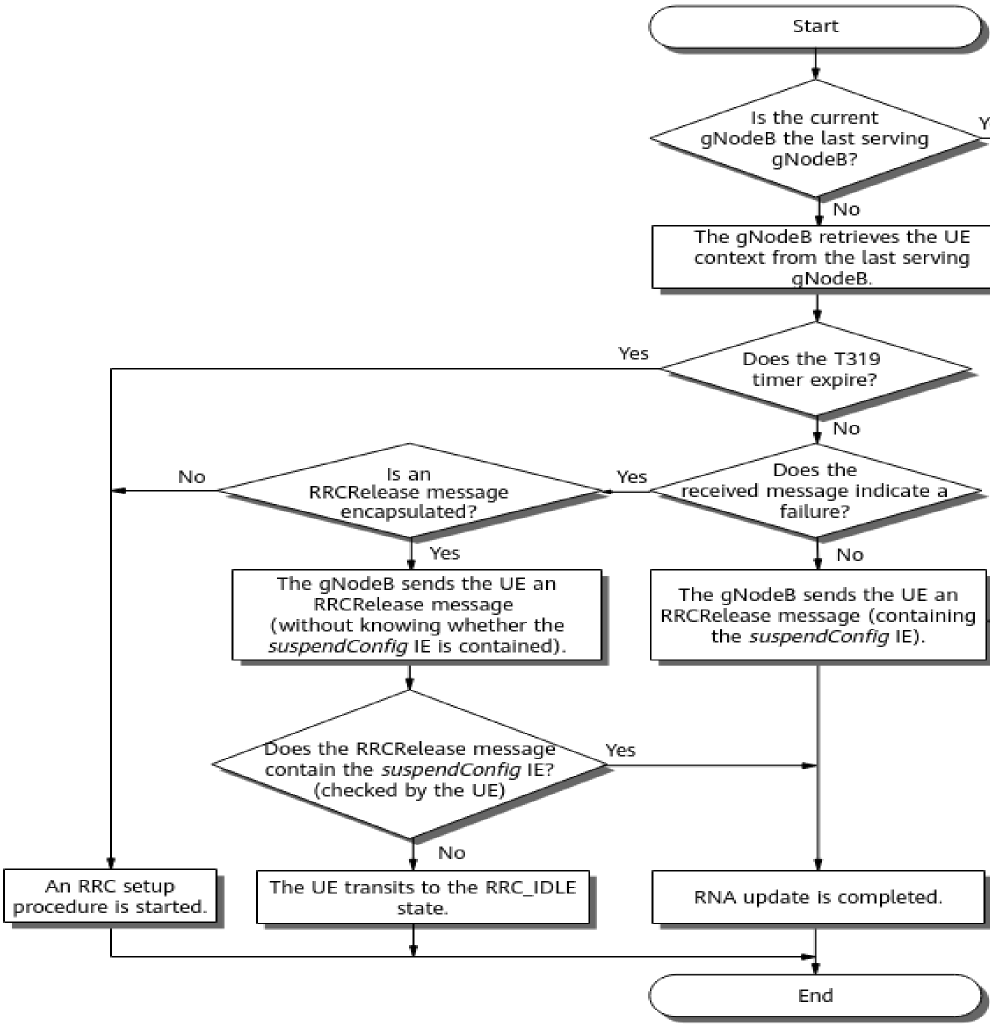
4.1.2 RNA Update

RNA update helps acquire the correct RNA of a moving UE. RNA update is triggered in any of the following scenarios:

- **A UE periodically sends an RRCResumeRequest1 or RRCResumeRequest message with the "Resume Cause" being "RNA Update" to the gNodeB. This enables the gNodeB to determine whether the UE is disconnected from the network.**
- **Once the UE discovers that the RNA ID of the current cell after cell reselection differs from the last-acquired RNA ID, it notifies the gNodeB by sending an RRCResumeRequest1 or RRCResumeRequest message with the "Resume Cause" being "RNA Update".**

After receiving an RRCResumeRequest1 or RRCResumeRequest message with the "Resume Cause" being "RNA Update" from the UE, the gNodeB starts an RNA update procedure as illustrated in **Figure 4-3**.

Figure 4-3 RNA update procedure



RNA update is classified into intra-gNodeB RNA update and inter-gNodeB RNA update, depending on whether the gNodeB currently serving the UE is the last serving gNodeB.

- In the case of intra-gNodeB RNA update (when the current gNodeB is the last serving gNodeB), the gNodeB sends the UE an RRCRelease message containing the *suspendConfig* IE to complete the RNA update.
- In the case of inter-gNodeB RNA update (when the current gNodeB is not the last serving gNodeB), the gNodeB sends a RETRIEVE UE CONTEXT REQUEST message to the last serving gNodeB over the Xn interface to retrieve the UE context. For details, see **4.1.4 UE Context Retrieval**.
  - If the gNodeB successfully retrieves the UE context information from the last serving gNodeB before the T319 timer expires, the gNodeB sends the UE an RRCRelease message containing the *suspendConfig* IE to complete the RNA update.
  - If the gNodeB does not receive any feedback from the last serving gNodeB before the T319 timer expires, the gNodeB sends an RRCSetup message to the UE.

– If the gNodeB receives a RETRIEVE UE CONTEXT FAILURE message from the last serving gNodeB before the T319 timer expires, it performs as follows:

  ■ If the message contains an encapsulated RRCRelease message, the gNodeB sends the RRCRelease message to the UE. However, the gNodeB does not know whether the encapsulated RRCRelease message contains the *suspendConfig* IE. Based on whether the received RRCRelease message contains the *suspendConfig* IE, the UE determines whether to transit to the RRC_IDLE state, or stay in the RRC_INACTIVE state and complete the RNA update.

  ■ If the message does not contain an encapsulated RRCRelease message, the gNodeB sends an RRCSetup message to the UE.

The length of the T319 timer is specified by the **NRDUCellUeTimerConst**.***T319*** parameter.

NOTE

During an inter-gNodeB RNA update, the RETRIEVE UE CONTEXT REQUEST message sent over the Xn interface carries the *PLMN*, *gNBID*, *CellId*, *PCI*, and *DL ARFCN* IEs corresponding to the target cell, in accordance with a Huawei proprietary protocol. These IEs help the last serving gNodeB derive keys.

If a RAN paging procedure is triggered due to the arrival of downlink data transmission and an inter-gNodeB RNA update procedure is also triggered for the UE in the RRC_INACTIVE state, then the last serving gNodeB sends a RETRIEVE UE CONTEXT RESPONSE message to the current gNodeB of the UE, notifying that downlink data transmission arrives. Upon receipt of this message, the current gNodeB enables the UE to transit from the RRC_INACTIVE state to the RRC_CONNECTED state for downlink data transmission.

For details about the signaling procedure for RNA update, see section 9.2.2.5 "RNA update" in 3GPP TS 38.300.

4.1.3 State Transition from RRC_INACTIVE

An RRC_INACTIVE UE can transit to the RRC_CONNECTED or RRC_IDLE state.

- Transition to RRC_CONNECTED: When a UE receives a RAN paging message or has data to send to the network, the UE transits from RRC_INACTIVE to RRC_CONNECTED. For details, see section 9.2.2.4 "State Transitions" in 3GPP TS 38.300 V15.6.0.
- Transition to RRC_IDLE:
  – If a UE in the RRC_INACTIVE state does not perform any services throughout the period specified by **gNBConnStateTimer**.***UeInactiveToIdleTimer*** (UE inactive-to-idle timer) and the serving gNodeB remains unchanged, the last serving gNodeB triggers the UE to transit from RRC_INACTIVE to RRC_IDLE and sends a context release message to the core network.

    RRC_INACTIVE UEs also occupy the capacity for the maximum number of RRC_CONNECTED UEs supported by a base station, as the gNodeB still stores the historical contexts of RRC_INACTIVE UEs. Against this background, the load optimization based on the number of RRC_INACTIVE UEs function can be used to minimize the impact of RRC_INACTIVE UEs that occupy the capacity for the maximum number of

RRC_CONNECTED UEs supported by a base station. This function is controlled by the **INACTIVE_UE_NUM_LOAD_OPT_SW** option of the **gNodeBParam**.***BoardProcessingPolicySw*** parameter.

When load optimization based on the number of RRC_INACTIVE UEs is enabled, the implementation is as follows:

- **In the case that the number of RRC_INACTIVE UEs on the main control board is less than or equal to 80% of the maximum number of RRC_INACTIVE UEs supported:**

  The gNodeB uses the configured length as the actual length of the UE inactive-to-idle timer.

- **In the case that the number of RRC_INACTIVE UEs on the main control board exceeds 80% of the maximum number of RRC_INACTIVE UEs supported:**

  The gNodeB starts to step down the actual length of the UE inactive-to-idle timer, so that RRC_INACTIVE UEs can transit to the RRC_IDLE state faster. Specifically, the actual length of the UE inactive-to-idle timer is reduced by a step specified by **gNBConnStateTimer**.***InactiveToIdleTimerAdjStep*** each time the number of RRC_INACTIVE UEs increases by 1% of the maximum number of RRC_INACTIVE UEs supported.

– If RAN paging fails due to reasons such as no available Xn connection to the last serving gNodeB, the gNodeB triggers the UE to transit to the RRC_IDLE state after the RAN paging timer (**gNBUuInteraction**.***RanPagingTimer***) expires and waits for the core network to initiate 5GC paging.

NOTE

During an inter-gNodeB state transition for an RRC_INACTIVE UE, the RETRIEVE UE CONTEXT REQUEST message sent over the Xn interface carries the *PLMN*, *gNBID*, *CellId*, *PCI*, and *DL ARFCN* IEs corresponding to the target cell, in accordance with a Huawei proprietary protocol. These IEs help the last serving gNodeB derive keys.

4.1.4 UE Context Retrieval

During an inter-gNodeB RNA update or transition from RRC_INACTIVE to RRC_CONNECTED, the target gNodeB needs to retrieve UE context information from the last serving gNodeB over the Xn interface as follows:

1. When a UE transits from RRC_CONNECTED to RRC_INACTIVE, the last serving gNodeB sends an RRCRelease message containing a full I-RNTI and a short I-RNTI to the UE. The full I-RNTI consists of the 20 least significant bits of the gNodeB ID of the last serving gNodeB and a temporarily generated UE ID. The short I-RNTI consists of the 12 least significant bits of the gNodeB ID of the last serving gNodeB and a temporarily generated UE ID. I-RNTI is short for inactive radio network temporary identifier.
2. When the UE sends the target gNodeB an RRCResumeRequest1 or RRCResumeRequest message containing the I-RNTI allocated by the source gNodeB, the target gNodeB resolves the I-RNTI to obtain the 20 or 12 least significant bits of the gNodeB ID of the last serving gNodeB. Then the target gNodeB sends a RETRIEVE UE CONTEXT REQUEST message to the last serving gNodeB over the Xn interface to retrieve UE context information.

To ensure that the target gNodeB can send the request to the correct last serving gNodeB, the 20 or 12 least significant bits of each gNodeB ID must be unique in the contiguous coverage areas. If the 20 or 12 least significant bits of a gNodeB ID are not unique, the target gNodeB may find and send Retrieve UE Context Request messages to multiple gNodeBs meeting the requirements, causing context retrieval failures.

4.2 Network Analysis

4.2.1 Benefits

UEs in the RRC_INACTIVE state can maintain the similar level of power consumption as they did in the RRC_IDLE state while resuming data transmission after a short delay because less signaling is exchanged over the air interface and with the core network. The specific delay varies with factors such as UE processing and core network capabilities.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

After the RRC_INACTIVE state is enabled, the number of messages over the Xn interface may increase, mainly caused by RAN paging of UEs in the RRC_INACTIVE state. When the UE inactivity timer expires, UEs enter the RRC_INACTIVE state and, in this case, their QoS flows are not released. As such, the number of normal QoS flow releases decreases and the QoS flow service drop rate (**Service Call Drop Rate (CU, Inactive)**) increases.

After the RRC_INACTIVE state is enabled, the UE model over the Uu interface is noticeably changed, as a UE transits from the RRC_INACTIVE state to the RRC_CONNECTED state through an RRC resume procedure whereas the UE transits from the RRC_IDLE state to the RRC_CONNECTED state through an RRC setup procedure. As such, the related impacts are as follows:

- The signaling over the Uu interface changes, which is mainly caused by state transitions and periodic RNA updates.
- The number of paging messages over the Uu interface increases when RRC_INACTIVE UEs are paged for downlink data transmission.
- The numbers of security mode setups and RRC connection reconfigurations decrease, whereas the number of RRC connection resume times increases.
- The number of RRC connection setup requests increases. In addition, Msg3 and Msg4 during an RRC resume procedure contain more information and come with larger sizes than Msg3 and Msg4 during an RRC setup procedure. As such, the probability of message sending and receiving failures increases, resulting in a decrease in **RRC Setup Success Rate (CU, Inactive)** and **QoS Flow Setup Success Rate (CU, Inactive)**.
- In a transition from the RRC_IDLE state to the RRC_CONNECTED state, DRBs are established only after the procedures such as RRC reconfiguration, security mode setup, context setup, and authentication are complete. In contrast, a transition from the RRC_INACTIVE state to the RRC_CONNECTED state

requires only an RRC resume procedure, meaning that the aforementioned procedures are not required. This significantly reduces signaling exchanges and accelerates the establishment of DRBs. As such, the number of DRBs in the cell may increase.

- After the RRC_INACTIVE state is enabled, UEs that are originally arranged to transit from the RRC_CONNECTED state to the RRC_IDLE state will transit to the RRC_INACTIVE state, if they support the RRC_INACTIVE state. The contexts, QoS flows, and PDU sessions of UEs that transit to the RRC_INACTIVE state are suspended. As such, the numbers of normal UE context releases, normal QoS flow releases, and normal PDU session releases in the cell decrease.
- As RRCResume and RRCReconfiguration messages are scheduled differently, the values of the following indicators may be affected:
  - Impact on accessibility-related indicators
    - The contention-based access success rate fluctuates. Specifically, the values of the following indicators fluctuate:
      - **N.RA.Contention.Att**
      - **N.RA.Contention.Resp**
      - **N.RA.Contention.Msg3**
      - **N.RA.Contention.Resolution.Succ**
      - **N.RA.SDT.Contention.Att**
      - **N.RA.SDT.Contention.Resp**
      - **N.RA.Contention.Att.Max**
    - The non-contention-based access success rate fluctuates. Specifically, the values of the following indicators fluctuate:
      - **N.RA.Dedicated.Att**
      - **N.RA.Dedicated.Ho.Att**
      - **N.RA.Dedicated.Resp**
      - **N.RA.Dedicated.Msg3**
  - Impact on scheduling-related indicators
    - The average uplink and downlink ranks fluctuate.
      - Average downlink rank = (1 x **N.PDSCH.InitTbDl.Rank1** + 2 x **N.PDSCH.InitTbDl.Rank2** + 3 x **N.PDSCH.InitTbDl.Rank3** + 4 x **N.PDSCH.InitTbDl.Rank4**)/(**N.PDSCH.InitTbDl.Rank1** + **N.PDSCH.InitTbDl.Rank2** + **N.PDSCH.InitTbDl.Rank3** + **N.PDSCH.InitTbDl.Rank4**)
      - Average uplink rank = (1 x **N.PUSCH.InitTbUl.Rank1** + 2 x **N.PUSCH.InitTbUl.Rank2**)/(**N.PUSCH.InitTbUl.Rank1** + **N.PUSCH.InitTbUl.Rank2**)
    - The average uplink and downlink MCS indexes fluctuate.
      - Average downlink MCS index = ∑(*n* x N.ChMeas.PDSCH.MCS.*n*)/∑(N.ChMeas.PDSCH.MCS.*n*), where *n* ranges from 0 to 31 (for example, **N.ChMeas.PDSCH.MCS.1**)

    - Average uplink MCS index = ∑(*n* x N.ChMeas.PUSCH.MCS.*n*)/∑(N.ChMeas.PUSCH.MCS.*n*), where *n* ranges from 0 to 31 (for example, **N.ChMeas.PUSCH.MCS.1**)

  - The uplink and downlink IBLERs fluctuate.
    - Downlink IBLER = (**N.DL.SCH.QPSK.ErrTB.Ibler** + **N.DL.SCH.16QAM.ErrTB.Ibler** + **N.DL.SCH.64QAM.ErrTB.Ibler** + **N.DL.SCH.256QAM.ErrTB.Ibler** + **N.DL.SCH.1024QAM.ErrTB.Ibler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB** + **N.DL.SCH.1024QAM.TB**) x 100%
    - Uplink IBLER = (**N.UL.SCH.HalfPiBPSK.ErrTB.Ibler** + **N.UL.SCH.QPSK.ErrTB.Ibler** + **N.UL.SCH.16QAM.ErrTB.Ibler** + **N.UL.SCH.64QAM.ErrTB.Ibler** + **N.UL.SCH.256QAM.ErrTB.Ibler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%

  - The uplink and downlink RBLERs fluctuate.
    - Downlink RBLER = (**N.DL.SCH.QPSK.ErrTB.Rbler** + **N.DL.SCH.16QAM.ErrTB.Rbler** + **N.DL.SCH.64QAM.ErrTB.Rbler** + **N.DL.SCH.256QAM.ErrTB.Rbler** + **N.DL.SCH.1024QAM.ErrTB.Rbler**)/(**N.DL.SCH.QPSK.TB** + **N.DL.SCH.16QAM.TB** + **N.DL.SCH.64QAM.TB** + **N.DL.SCH.256QAM.TB** + **N.DL.SCH.1024QAM.TB**) x 100%
    - Uplink RBLER = (**N.UL.SCH.HalfPiBPSK.ErrTB.Rbler** + **N.UL.SCH.QPSK.ErrTB.Rbler** + **N.UL.SCH.16QAM.ErrTB.Rbler** + **N.UL.SCH.64QAM.ErrTB.Rbler** + **N.UL.SCH.256QAM.ErrTB.Rbler**)/(**N.UL.SCH.HalfPiBPSK.TB** + **N.UL.SCH.QPSK.TB** + **N.UL.SCH.16QAM.TB** + **N.UL.SCH.64QAM.TB** + **N.UL.SCH.256QAM.TB**) x 100%

- Impact on MU-MIMO indicators

  - The average number of paired layers for uplink and downlink MU pairing fluctuates.

    Average number of paired layers for downlink MU pairing = **N.ChMeas.MIMO.DL.Pair.Layer**/**N.ChMeas.MIMO.DL.Pair.PRB**

    Average number of paired layers for uplink MU pairing = **N.ChMeas.MIMO.UL.Pair.Layer**/**N.ChMeas.MIMO.UL.Pair.PRB**

  - The average proportion of PRBs for uplink and downlink MU pairing fluctuates.

    Average proportion of PRBs for downlink MU pairing = **N.ChMeas.MIMO.DL.Pair.PRB**/(**N.PRB.DL.Used.Avg** x **N.DL.PDSCH.Tti.Num**) x 100%

    Average proportion of PRBs for uplink MU pairing = **N.ChMeas.MIMO.UL.Pair.PRB**/(**N.PRB.UL.Used.Avg** x **N.UL.PUSCH.Tti.Num**) x 100%

- Impact on coverage-related indicators

  The average uplink and downlink CCE aggregation levels fluctuate.

- Average downlink CCE aggregation level = (**N.CCE.DL.AggLvl16Num** x 16 + **N.CCE.DL.AggLvl8Num** x 8 + **N.CCE.DL.AggLvl4Num** x 4 + **N.CCE.DL.AggLvl2Num** x 2 + **N.CCE.DL.AggLvl1Num**)/(**N.CCE.DL.AggLvl16Num** + **N.CCE.DL.AggLvl8Num** + **N.CCE.DL.AggLvl4Num** + **N.CCE.DL.AggLvl2Num** + **N.CCE.DL.AggLvl1Num**)
- Average uplink CCE aggregation level = (**N.CCE.UL.AggLvl16Num** x 16 + **N.CCE.UL.AggLvl8Num** x 8 + **N.CCE.UL.AggLvl4Num** x 4 + **N.CCE.UL.AggLvl2Num** x 2 + **N.CCE.UL.AggLvl1Num**)/(**N.CCE.UL.AggLvl16Num** + **N.CCE.UL.AggLvl8Num** + **N.CCE.UL.AggLvl4Num** + **N.CCE.UL.AggLvl2Num** + **N.CCE.UL.AggLvl1Num**)

– Impact on delay-related indicators

- The processing delay at the MAC layer in the gNodeB fluctuates.

  Processing delay at the MAC layer in the gNodeB = (**N.QoS.DL.PktDelayAirInterface.Time**/**N.QoS.DL.PktDelayAirInterface.Num**)/1000
- The processing delay at the RLC layer in the gNodeB fluctuates.

  Processing delay at the RLC layer in the gNodeB = (**N.QoS.DL.PktDelaygNBDU.Time**/**N.QoS.DL.PktDelaygNBDU.Num**)/1000
- The first packet delay (**N.Traffic.DL.RlcFirstPktDelay.Time**) fluctuates.
- After UEs enter the RRC_INACTIVE state, packets of established bearers are buffered at the PDCP layer and transmitted after the bearers are reactivated. Therefore, the average downlink delay of 5QI-specific services at the PDCP layer increases.

  Average downlink delay of 5QI-specific services at the PDCP layer = **N.PDCP.DL.Cell5QI.Delay**/**N.PDCP.DL.Packets.Cell5QI**

– Impact on load-related indicators

The uplink and downlink PRB usage fluctuate.

- Downlink PRB usage = (**N.PRB.DL.Used.Avg**/**N.PRB.DL.Avail.Avg**) x 100%
- Uplink PRB usage = (**N.PRB.UL.Used.Avg**/**N.PRB.UL.Avail.Avg**) x 100%

– Impact on indicators related to the traffic volume

- The average uplink cell throughput (indicated by **Cell Uplink Average Throughput (DU)**) and average downlink cell throughput (indicated by **Cell Downlink Average Throughput (DU)**) fluctuate.
- The average uplink UE throughput (indicated by **User Uplink Average Throughput (DU)**) and average downlink UE throughput (indicated by **User Downlink Average Throughput (DU)**) fluctuate.

When the gNodeB sends an RRC SETUP message to a UE in the RRC_INACTIVE state for RRC connection setup and the core network sends a UE CONTEXT

RELEASE COMMAND message to the last serving gNodeB of the UE, if the cause value contained in the latter message is not "UE context transfer", the last serving gNodeB cannot be aware that the RRC connection has been reestablished for the UE and therefore still initiates an invalid RAN paging to the UE. As a result, the number of RAN-initiated paging requests in the source cell (**N.Paging.RAN.Src.Att**) of the last serving gNodeB increases and the number of successful RAN-initiated pagings in the source cell (**N.Paging.RAN.Src.Succ**) of the last serving gNodeB decreases.

4.3 Requirements

4.3.1 Licenses

There are no license requirements.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

None

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast RRC connection release	**RRC_RELEASE_QUICK_SW** option of the **NRCellAlgoSwitch**.***UePwrSavingSwitch*** parameter	*Device-Pipe Synergy*	When the RRC_INACTIVE state function takes effect, fast RRC connection release allows UEs to quickly switch from the RRC_CONNECTED state to the RRC_INACTIVE state.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Fast RRC connection release	**UE_RELEASE_PREFERENCE_SW** option of the **NRCellAlgoSwitch**.***UePwrSavingSwitch*** parameter	*UE Power Saving*	When the RRC_INACTIVE state function takes effect, fast RRC connection release allows UEs to quickly switch from the RRC_CONNECTED state to the RRC_INACTIVE state.
High-frequency TDD	None	None	None	None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

- UE requirements:

  UEs for which SUL takes effect do not enter the RRC_INACTIVE state due to restrictions defined in 3GPP specifications.

  UEs must support the RRC_INACTIVE state. The *inactiveState* IE in the UECapabilityInformation message over the Uu interface indicates support for the state.

- Networking requirements:
  - An Xn interface must be set up between gNodeBs in an RNA and between gNodeBs in adjacent RNAs so that the target gNodeB can retrieve context information from the last serving gNodeB over the Xn interface during an RNA update.
  - gNodeB IDs of long lengths, if required, must be properly planned on the live network so that the least significant 20 bits of each gNodeB ID are unique in contiguous coverage areas.
  - The gNodeBs in an RNA must belong to the same TA.
  - The target gNodeB must meet the following requirements for obtaining UE context:

    The target gNodeB must comply with 3GPP TS 38.423 V15.5.0 or later. Otherwise, UE context will fail to be obtained as the last serving gNodeB complies with 3GPP TS 38.423 V15.5.0 or later. Consequently, the function of protocol compatibility processing for Service Area Information exchanged between base stations has been introduced. This function is controlled by the **XN_SERVICE_AREA_INFO_PROC_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter. It is recommended that this option be deselected when different base stations comply with different protocol versions.

    - If this option is selected, Huawei base stations send and receive the *Service Area Information* IE in compliance with 3GPP TS 38.423 V15.5.0 or later.
    - If this option is deselected, Huawei base stations do not send the *Service Area Information* IE and they receive the IE in compliance with 3GPP TS 38.423 V15.4.0 or earlier. With this setting, the target gNodeB can obtain the UE context, but in the case of handovers initiated after transitions to the RRC_CONNECTED state, UEs may be handed over to the forbidden cells indicated by the *Mobility Restriction List* IE in the INITIAL CONTEXT SETUP REQUEST message sent from the AMF.
  - Compared with 3GPP TS 38.423 V16.6.0, 3GPP TS 38.423 V16.7.0 introduced an incompatibility change in the definition of the *UE Security Capabilities* IE. For details, see section 9.2.3.49 "UE Security Capabilities" in 3GPP TS 38.423 V16.7.0. During an RNA update, the gNodeB processes the *UE Security Capabilities* IE in the Retrieve UE Context Response message sent by the last serving gNodeB to select a ciphering algorithm. If the two base stations comply with different protocols during the processing, a protocol compatibility issue occurs. To prevent service restoration failures due to protocol inconsistency between the gNodeB and the last serving gNodeB, Huawei gNodeBs have optimized the compatibility processing in accordance with 3GPP TS 38.423 V16.7.0. This optimization is controlled by the **XN_UE_SECURITY_CAP_PROC_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter. It is recommended that this option be selected when a Huawei gNodeB is connected to a non-Huawei gNodeB over the Xn interface and the non-Huawei gNodeB sends and receives the Retrieve UE Context Response message containing the *UE Security Capabilities* IE in compliance with 3GPP TS 38.423 V16.7.0.

- When this option is selected, Huawei gNodeBs send and receive the *UE Security Capabilities* IE in compliance with 3GPP TS 38.423 V16.7.0.
- When this option is deselected, Huawei gNodeBs send and receive the *UE Security Capabilities* IE in compliance with 3GPP TS 38.423 V16.6.0.

- Core network requirements: The core network must support the RRC_INACTIVE state. The support is indicated by the presence of the *Core Network Assistance Information for RRC INACTIVE* IE in the INITIAL CONTEXT SETUP REQUEST, UE CONTEXT MODIFICATION REQUEST, HANDOVER REQUEST, or PATH SWITCH REQUEST ACKNOWLEDGE message over the NG interface, and this IE does not contain the *MICO Mode Indication* sub-IE.

4.4 Operation and Maintenance

4.4.1 Data Configuration

Activating this function requires the planning of RNAs and neighboring cells. That is, RNA IDs and neighbor relationships should be configured before this function is activated. For details about how to configure neighbor relationships, see *SA Mobility Management in Connected Mode*. If ANR is enabled, neighboring cells do not require manual configuration. For details, see *ANR*.

4.4.1.1 Data Preparation

Table 4-1 and **Table 4-2** describe the parameters used for function activation and optimization, respectively.

Table 4-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
RAN Notification Area ID	**NRDUCell.*RanNotificationAreaId***	Set this parameter based on the network plan.
RAN Notification Area ID	**NRDUCellOp.*RanNotificationAreaId***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
INACTIVE Strategy Switch	**NRCellAlgoSwitch**.***InactiveStrategySwitch***	Mobility management in inactive mode is controlled by the **RRC_INACTIVE_SWITCH** option. If this option is selected, UEs in inactive mode can maintain a similar level of power consumption as in idle mode while resuming data transmission within a short delay. You are advised to select the **RRC_INACTIVE_SWITCH** option in delay-sensitive scenarios where there are UEs in inactive mode in a cell.

NOTE

An RNA ID consists of a TAC and a RAN area code.

In scenarios without multi-operator sharing, the RAN area code is specified by the **NRDUCell**.***RanNotificationAreaId*** parameter. In multi-operator sharing scenarios, the operator-specific RAN area code is specified by the **NRDUCellOp**.***RanNotificationAreaId*** parameter. If cell-list-based automatic RNA planning has been enabled, manual configuration of the RAN area code is not required. For details about cell-list-based automatic RNA planning, see *UE Power Saving*.

Table 4-2 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
UE Inactivity Timer	**NRDUCellQciBearer**.***UeInactivityTimer***	Set this parameter based on the network plan.
Timer 319	**NRDUCellUeTimerConst**.***T319***	Use the recommended value.
UE Inactive To Idle Timer	**gNBConnStateTimer**.***UeInactiveToIdleTimer***	Use the recommended value.
Board Processing Policy Switch	**gNodeBParam**.***BoardProcessingPolicySw***	When the main control board serves a large number of RRC_INACTIVE UEs, you can select the **INACTIVE_UE_NUM_LOAD_OPT_SW** option to minimize the impact of RRC_INACTIVE UEs that occupy the capacity for the maximum number of RRC_CONNECTED UEs supported by a base station.

Parameter Name	Parameter ID	Setting Notes
UE Inactive-to-Idle Timer Adjust Step	**gNBConnStateTimer**.***InactiveToIdleTimerAdjStep***	Use the recommended value.
Compatibility Algorithm Switch	**gNodeBParam**.***CompatibilityAlgoSwitch***	It is recommended that the **XN_SERVICE_AREA_INFO_PROC_SW** option be deselected when the source and target gNodeBs comply with different versions of 3GPP TS 38.423 (one complying with V15.5.0 or later whereas the other complying with V15.4.0 or earlier). Otherwise, it is recommended that the **XN_SERVICE_AREA_INFO_PROC_SW** option be selected.
Compatibility Algorithm Switch	**gNodeBParam**.***CompatibilityAlgoSwitch***	It is recommended that the **XN_UE_SECURITY_CAP_PROC_SW** option be selected when a Huawei gNodeB is connected to a non-Huawei gNodeB over the Xn interface and the non-Huawei gNodeB sends and receives the Handover Request message containing the *UE Security Capabilities* IE in compliance with 3GPP TS 38.423 V16.7.0. In other scenarios, it is recommended that the **XN_UE_SECURITY_CAP_PROC_SW** option be deselected.
RAN Paging Timer	**gNBUuInteraction**.***RanPagingTimer***	Use the recommended value.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

TDD configuration examples

```
//Configuring RNA IDs for local cells (without operator-specific configuration)
MOD NRDUCELL: NrDuCellId=0, DuplexMode=CELL_TDD, CellId=0, TrackingAreaId=0,
RanNotificationAreaId=1;
MOD NRDUCELL: NrDuCellId=1, DuplexMode=CELL_TDD, CellId=1, TrackingAreaId=0,
RanNotificationAreaId=1;
//Configuring RNA IDs for local cells (with operator-specific configuration)
MOD NRDUCELLOP: NrDuCellId=0, OperatorId=0, RanNotificationAreaId=1;
MOD NRDUCELLOP: NrDuCellId=0, OperatorId=1, RanNotificationAreaId=2;
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=0, RanNotificationAreaId=1;
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=1, RanNotificationAreaId=2;
//(Optional, required when no neighbor relationship has been configured) Adding a neighbor relationship
with an NR cell
ADD NRCELLRELATION: NrCellId=0, Mcc="302", Mnc="220", gNBId=1, CellId=0;
//Turning on the RRC_INACTIVE state switch
MOD NRCELLALGOSWITCH: NrCellId=0, InactiveStrategySwitch=RRC_INACTIVE_SWITCH-1;
```

Optimization Command Examples

TDD configuration examples

```
//Changing the UE inactivity timer length to 10s, which should be adjusted based on the network plan
MOD NRDUCELLQCIBEARER: NrDuCellId=0, Qci=9, UeInactivityTimer=10;
//Changing the T319 timer length to 1s
MOD NRDUCELLUETIMERCONST: NrDuCellId=0, T319=MS1000;
//Changing the UE inactive-to-idle timer length to 1500s
MOD GNBCONNSTATETIMER: UeInactiveToIdleTimer=1500;
//Turning on the switch for load optimization based on the number of RRC_INACTIVE UEs
MOD GNODEBPARAM: BoardProcessingPolicySw=INACTIVE_UE_NUM_LOAD_OPT_SW-1;
//Modifying the step by which the length of the UE inactive-to-idle timer is adjusted
MOD GNBCONNSTATETIMER: InactiveToIdleTimerAdjStep=60;
//Turning on the Xn service area information processing switch
MOD GNODEBPARAM: CompatibilityAlgoSwitch=XN_SERVICE_AREA_INFO_PROC_SW-1;
//Turning on the Xn UE security capability processing switch
MOD GNODEBPARAM: CompatibilityAlgoSwitch=XN_UE_SECURITY_CAP_PROC_SW-1;
//Modifying the RAN paging timer
MOD GNBUUINTERACTION: RanPagingTimer=50;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

TDD configuration examples

```
//Turning off the RRC_INACTIVE state switch
MOD NRCELLALGOSWITCH: NrCellId=0, InactiveStrategySwitch=RRC_INACTIVE_SWITCH-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Using Signaling Tracing

Feature activation observation:

Step 1 Start Uu signaling tracing for the cell that is enabled with the RRC_INACTIVE state function as follows: Log in to the MAE-Access and choose **Monitor** > **Signaling Trace** > **Signaling Trace Management**. On the displayed page, choose **Trace Type** > **NR** > **Application Layer** > **Uu Interface Trace**.

Step 2 Check for the *suspendConfig* IE in the RRC_REL messages of the tracing result. If the IE is found, the UE has entered the RRC_INACTIVE state and this function has taken effect.

Step 3 Check for the *resumeCause* IE in the RRC_RESUME_REQ_1 messages of the tracing result. If the IE with the value **rna-update** is found, the RNA has been updated.

----End

Service setup delay observation:

The service setup delay of a UE in the RRC_INACTIVE state is shorter than that in the RRC_IDLE state due to less signaling being exchanged over the air interface and with the core network.

Step 1 When a UE is in the RRC_INACTIVE state, observe its service setup duration from the time when the UE sends a preamble to the time when the UE sends an RRCResumeComplete message.

Step 2 When the UE is in the RRC_IDLE state, observe the service setup duration from the time when the UE sends a preamble to the time when the bearer setup is complete (indicated by the RRCReconfigurationComplete message).

----End

Using Counters

View the values of related counters to check the enabling of the RRC_INACTIVE state. If the value of any of the counters listed in **Table 4-3** is not 0, the RRC_INACTIVE state has been enabled.

Table 4-3 Counters related to the RRC_INACTIVE state

Counter ID	Counter Name
1911835994	**N.User.Inactive.Avg**
1911835995	**N.User.Inactive.Max**

4.4.3 Network Monitoring

This function can be monitored using the following KPIs:

- **RRC Setup Success Rate (CU, Inactive)**
- **QoS Flow Setup Success Rate (CU, Inactive)**
- **Service Call Drop Rate (CU, Inactive)**

5 Appendix

This section describes the signaling procedures for mobility management in SA networking. For details about the signaling procedures for mobility management in NSA networking, see *NSA Mobility Management*.

- SA mobility management in idle mode: Cell selection and reselection mainly occur on UEs and do not involve signaling exchange between UEs and the gNodeB.
- SA mobility management in connected mode: For details about the signaling procedures for intra-base-station, Xn-based, and NG-based handovers, see *SA Mobility Management in Connected Mode*.
- SA mobility management in inactive mode
  - For details about the signaling procedure for an RNA update, see **5.1 RNA Update**.
  - For details about the signaling procedure for a state transition from RRC_INACTIVE, see **5.2 State Transition from RRC_INACTIVE**.

5.1 RNA Update

RNA update helps acquire the correct RNA of a moving UE. RNA update is triggered in any of the following scenarios:

- A UE periodically sends an RRCResumeRequest1 or RRCResumeRequest message with the "Resume Cause" being "RNA Update" to the gNodeB. This enables the gNodeB to determine whether the UE is disconnected from the network.
- Once the UE discovers that the RNA ID of the current cell after cell reselection differs from the last-acquired RNA ID, it notifies the gNodeB by sending an RRCResumeRequest1 or RRCResumeRequest message with the "Resume Cause" being "RNA Update".

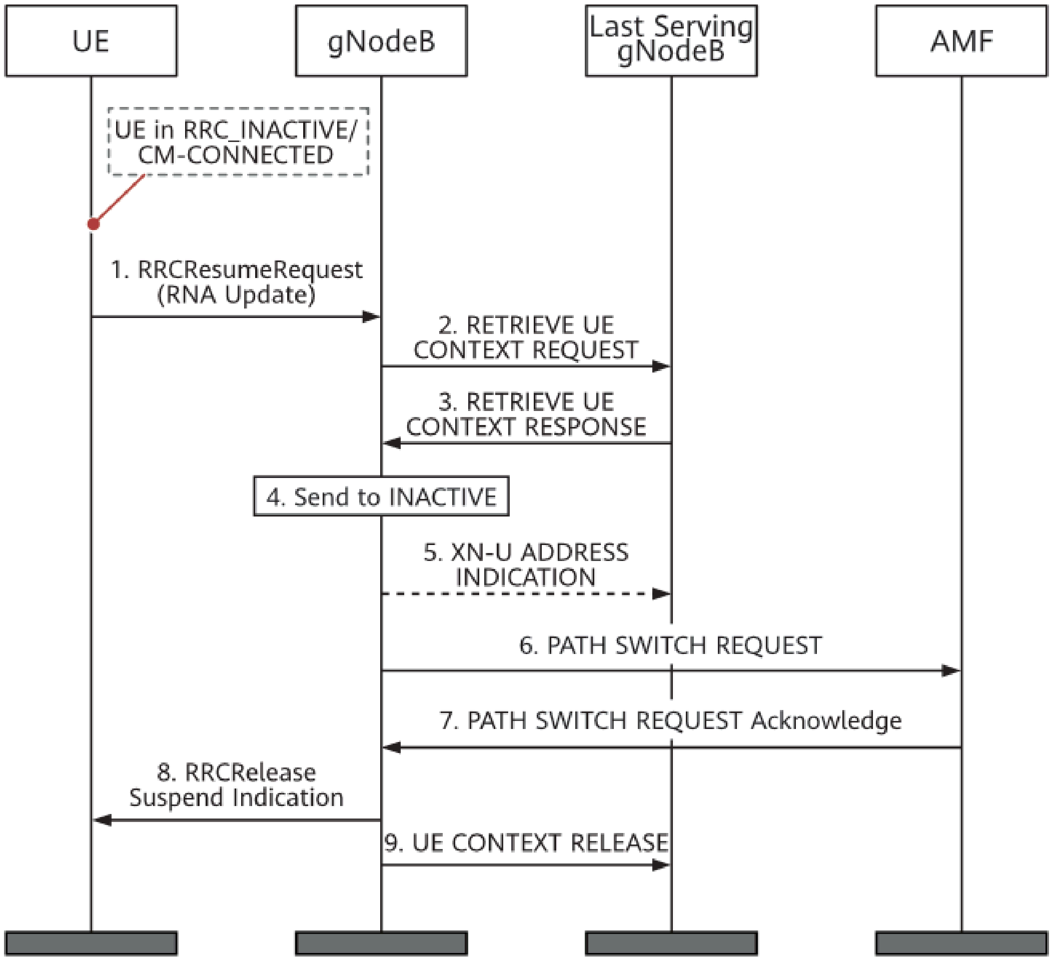
Upon receiving an RRCResumeRequest1 or RRCResumeRequest message with the "Resume Cause" being "RNA Update" from a UE, the gNodeB starts RNA update. RNA update can occur in the following scenarios:

- RNA update with UE context relocation

  When the UE moves out of the area covered by the last serving gNodeB, the gNodeB starts an RNA update procedure upon receipt of an

RRCResumeRequest1 or RRCResumeRequest message with the "Resume Cause" being "RNA Update" from the UE. In addition, the gNodeB requests the UE context from the last serving gNodeB, which responds to the gNodeB with the UE context. The gNodeB then manages the UE context. This process is referred to as UE context relocation. **Figure 5-1** illustrates the procedure for RNA update in this case.

Figure 5-1 Procedure for RNA update with UE context relocation



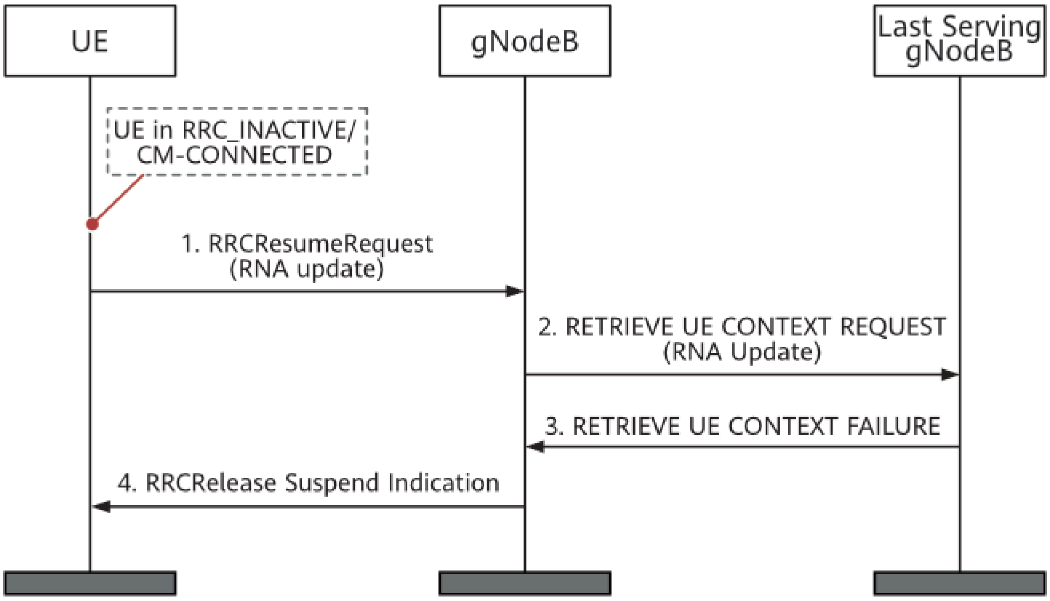
a. The UE resumes from the RRC_INACTIVE state, and provides the I-RNTI assigned by the last serving gNodeB and an appropriate cause value, for example, RNA update.
b. The gNodeB, if able to resolve the last serving gNodeB's identity contained in the I-RNTI, requests the last serving gNodeB to provide the UE context.
c. The last serving gNodeB provides the UE context.
d. The gNodeB instructs the UE to transit back to the RRC_INACTIVE state.
e. The gNodeB provides forwarding addresses if the loss of the downlink user data buffered in the last serving gNodeB should be prevented. That is, the gNodeB sends an XN-U ADDRESS INDICATION message to the last serving gNodeB.
f. The gNodeB performs path switching. That is, the gNodeB sends a PATH SWITCH REQUEST message to the AMF.

g. The AMF returns a PATH SWITCH REQUEST Acknowledge message to the gNodeB.
h. The gNodeB sends an RRCRelease message with a suspend indication to keep the UE in the RRC_INACTIVE state.
i. The gNodeB triggers the release of the UE resources at the last serving gNodeB.

- Periodic RNA update without UE context relocation

  The last serving gNodeB determines whether a UE has moved out of the configured RNA based on the I-RNTI. When the UE is still within the configured RNA, the last serving gNodeB does not provide the UE context to the gNodeB but continues to manage it. That is, no UE context relocation occurs. **Figure 5-2** illustrates the procedure for RNA update in this case.

Figure 5-2 Procedure for periodic RNA update without UE context relocation



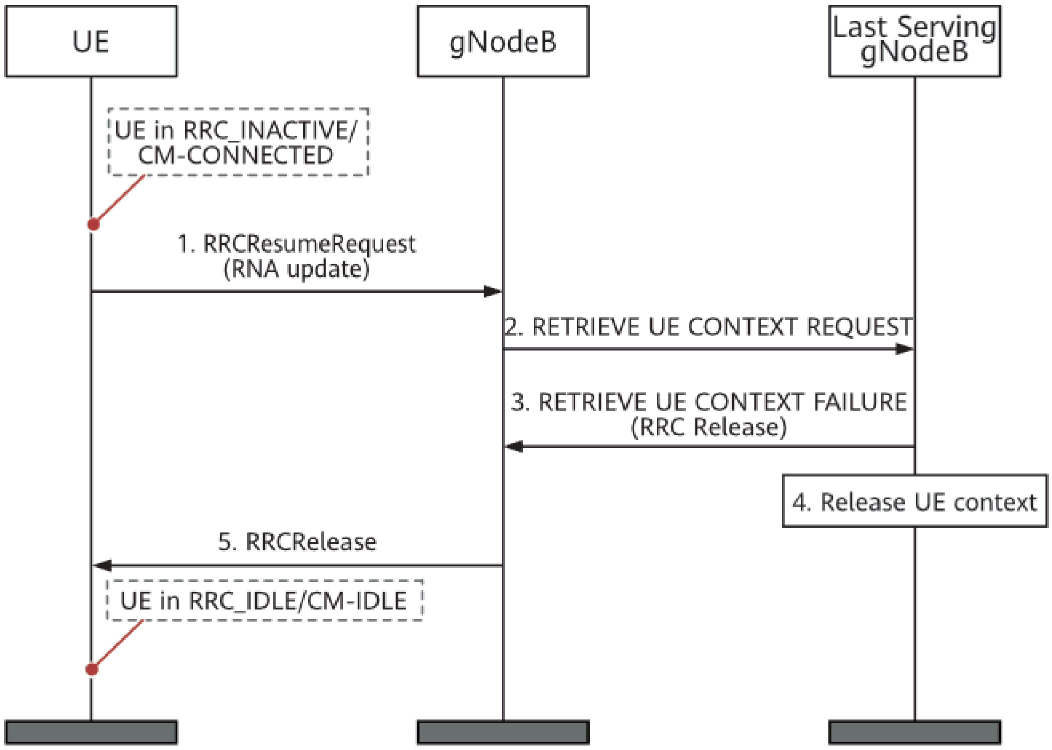
a. The UE resumes from the RRC_INACTIVE state, and provides the I-RNTI assigned by the last serving gNodeB and an appropriate cause value, for example, RNA update.
b. The gNodeB, if able to resolve the last serving gNodeB's identity contained in the I-RNTI, requests the last serving gNodeB to provide the UE context, providing the cause value received in step 1.
c. The last serving gNodeB stores received information (such as C-RNTI and PCI related to the resumption cell) for use in the next resume attempt, and responds to the gNodeB with a RETRIEVE UE CONTEXT FAILURE message that includes an encapsulated RRCRelease message. The RRCRelease message includes the *suspendConfig* IE.
d. The gNodeB forwards the RRCRelease message to the UE. The UE then updates the RNA based on the received RRCRelease message including the *suspendConfig* IE.

- RNA update with transition to the RRC_IDLE state

  If a UE in the RRC_INACTIVE state does not perform any service for a long time (specified by the UE inactive-to-idle timer

gNBConnStateTimer.***UeInactiveToIdleTimer***) and the serving gNodeB remains unchanged, the last serving gNodeB decides to push the UE to the RRC_IDLE state. **Figure 5-3** illustrates the procedure for RNA update in this case. Particularly, if RAN paging fails due to reasons such as no available Xn connection to the last serving gNodeB, the gNodeB also triggers the UE to transit to RRC_IDLE after the RAN paging timer (**gNBUuInteraction**.***RanPagingTimer***) expires. The UE then waits for 5GC paging. The procedure for RNA update in this case is the same as that illustrated in **Figure 5-3**.

Figure 5-3 Procedure for RNA update with transition to the RRC_IDLE state



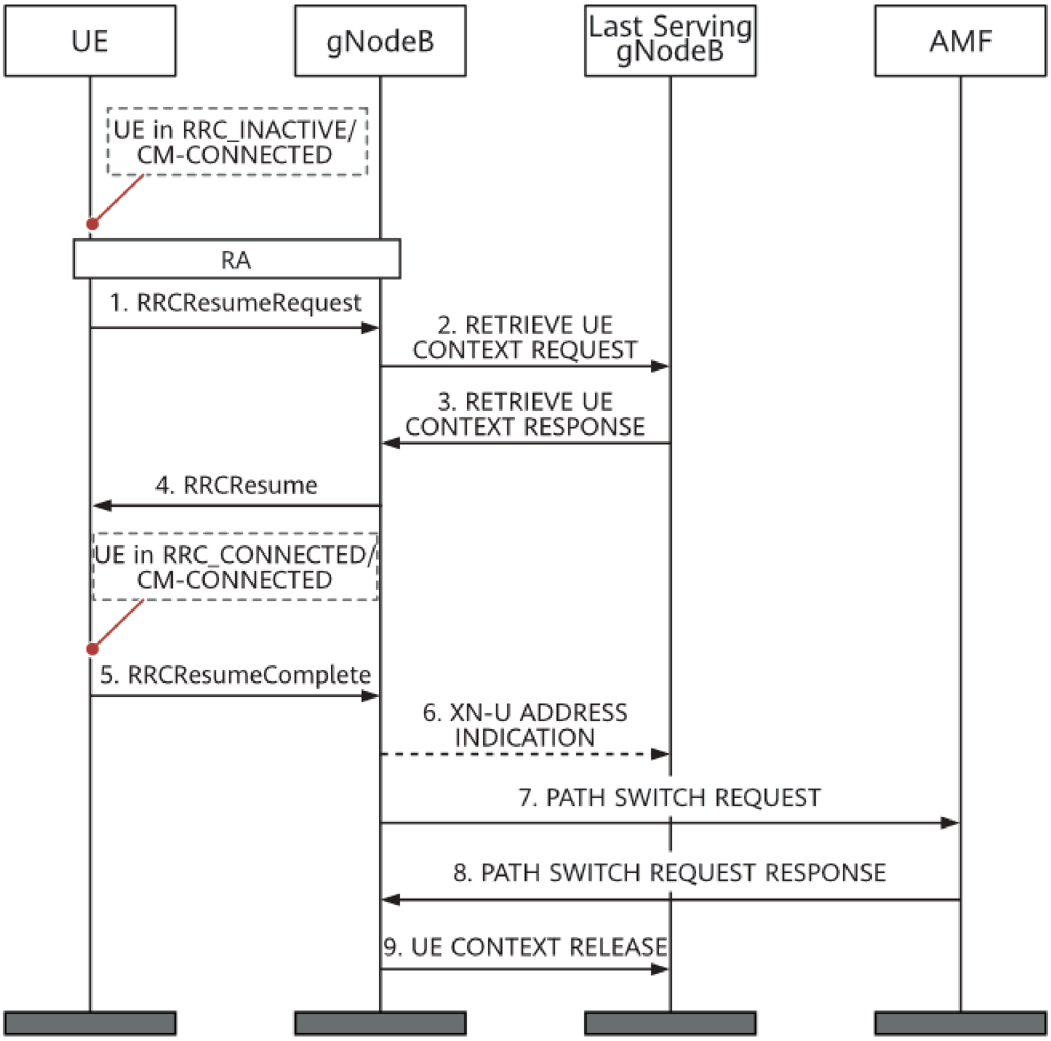
a. The UE resumes from the RRC_INACTIVE state, and provides the I-RNTI assigned by the last serving gNodeB and an appropriate cause value, for example, RNA update.
b. The gNodeB, if able to resolve the last serving gNodeB's identity contained in the I-RNTI, requests the last serving gNodeB to provide the UE context, providing the cause value received in step 1.
c. Instead of providing the UE context, the last serving gNodeB provides an RRCRelease message that does not include the *suspendConfig* IE to push the UE to the RRC_IDLE state.
d. The last serving gNodeB deletes the UE context.
e. The gNodeB sends an RRCRelease message (not including the *suspendConfig* IE) which triggers the UE to transit from the RRC_INACTIVE state to the RRC_IDLE state.

5.2 State Transition from RRC_INACTIVE

This section describes the signaling procedure for a state transition from RRC_INACTIVE to RRC_CONNECTED. For details about the signaling procedure for a state transition from RRC_INACTIVE to RRC_IDLE, see **Figure 5-3**.

Figure 5-4 illustrates the procedure for UE-triggered transition from RRC_INACTIVE to RRC_CONNECTED.

Figure 5-4 Procedure for UE-triggered transition from RRC_INACTIVE to RRC_CONNECTED

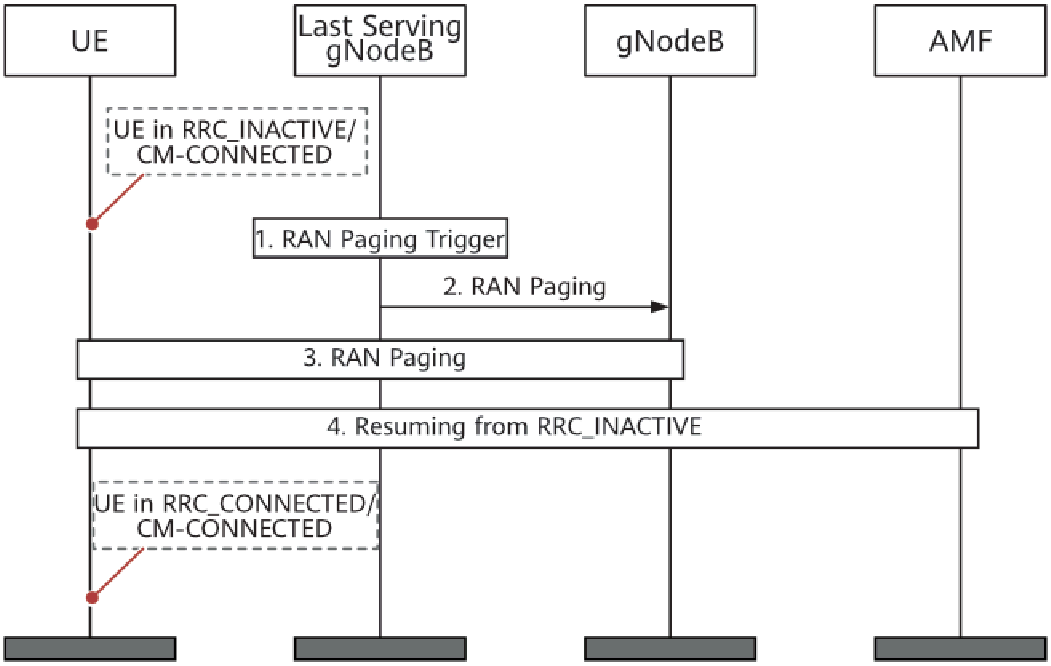


1. The UE resumes from the RRC_INACTIVE state and provides the I-RNTI assigned by the last serving gNodeB.
2. The gNodeB, if able to resolve the last serving gNodeB's identity contained in the I-RNTI, requests the last serving gNodeB to provide the UE context.
3. The last serving gNodeB provides the UE context.
4. The gNodeB instructs the UE to resume the RRC_CONNECTED state. That is, the gNodeB sends an RRCResume message to the UE.

5. The UE enters the RRC_CONNECTED state and returns an RRCResumeComplete message to the gNodeB.
6. The gNodeB provides forwarding addresses if the loss of the downlink user data buffered in the last serving gNodeB should be prevented. That is, the gNodeB sends an XN-U ADDRESS INDICATION message to the last serving gNodeB.
7. The gNodeB performs path switching. That is, the gNodeB sends a PATH SWITCH REQUEST message to the AMF.
8. The AMF returns a PATH SWITCH REQUEST RESPONSE message to the gNodeB.
9. The gNodeB triggers the release of the UE resources at the last serving gNodeB.

Figure 5-5 illustrates the procedure for network-triggered transition from RRC_INACTIVE to RRC_CONNECTED.

Figure 5-5 Procedure for network-triggered transition from RRC_INACTIVE to RRC_CONNECTED



1. A RAN paging-triggering event (such as incoming downlink user-plane data or downlink signaling from the 5GC) occurs.
2. RAN paging is triggered; either only in the cells served by the last serving gNodeB or also by means of Xn RAN paging in cells served by other gNodeBs, configured to the UE in the RNA.
3. The UE is paged based on the I-RNTI.
4. If the UE has been successfully reached, it attempts to resume from the RRC_INACTIVE state. For details about this process, see **Figure 5-4**.

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.423: "NG-RAN; Xn application protocol (XnAP)"
- *5G Networking and Signaling*
- *EPC-based NSA Basics*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *NSA-SA Selection Based on User Experience*
- *ANR*
- *Device-Pipe Synergy*
- *UE Power Saving*
- *NSA Mobility Management*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Idle Mode*
- *RedCap*
- *Idle Mode Management* in *eRAN Feature Documentation*
- *Mobility Management in Connected Mode* in *eRAN Feature Documentation*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*